

Walls or cascade? A pre-registered test of wall dissipation in decaying magnetohydrodynamic turbulence

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(Dated: 3 October 2026)

Does the small-scale dissipation of magnetohydrodynamic (MHD) turbulence come from the cascade in the bulk, as in a periodic box, or from the walls that confine laboratory plasmas and liquid metals? We test this with a controlled numerical experiment. The symmetries of the Taylor–Green MHD flow make the box $[0, \pi]^3$ a closed domain with free-slip, perfectly conducting walls: solving that box directly reproduces the periodic flow to round-off, and making its walls no-slip, with everything else unchanged, isolates what the walls do. With two independent lattice Boltzmann implementations we follow the decay at unit magnetic Prandtl number for Reynolds numbers 250 to 2000, on grids that resolve the $\text{Re}^{-1/2}$ wall layer, with the observable, the resolution criterion and the decision rule fixed before the runs. No-slip walls add dissipation within $\text{Re}^{-1/2}$ of them at every Reynolds number and under every definition we tried. The excess is viscous: the near-wall Ohmic dissipation is 40–45% lower than with free-slip walls. A second pre-registered test, on the energy budget, attributes that deficit to the stretching of the field by the flow, which the no-slip wall removes and, within $\text{Re}^{-1/2}$ of it, reverses. Four controls fixed in advance leave these findings standing, though one, a gentler start, enlarges the excess by up to 20% at $\text{Re} = 500$. Snapshots and three-dimensional views of the local dissipation show current sheets crossing the near-wall layer of the free-slip box and a broad viscous layer at the no-slip walls. The pre-registered reading, the excess at the dissipation peak, falls from 0.063 to 0.036 between $\text{Re} = 1000$ and 2000, the outcome we had called “cascade”. That fall is set by which of two maxima of the dissipation is the larger: the peak switches between rungs, and over a factor of eight in Reynolds number the trend of the wall share depends on when it is measured, while its sign does not.

I. INTRODUCTION

Decaying magnetohydrodynamic (MHD) turbulence has been studied mostly in periodic boxes.^{1–6} There the dissipation is made in current and vorticity sheets that the cascade forms in the bulk, and its rate tends to a value independent of the Reynolds number, Re ,^{2,5,6} the MHD counterpart of the finite dissipation of hydrodynamic turbulence.⁷

The flows that motivate much of this work, in liquid metals and laboratory plasmas, are bounded. In liquid-metal channels and ducts the magnetic Reynolds number is small and the field is imposed. Walls normal to it carry Hartmann layers, whose thickness falls as the inverse of the field,^{8,9} and direct simulations have mapped the turbulence next to those walls and to the walls parallel to the field.^{10–12} In a dynamo the flow generates the field, and the walls matter in another way. The conducting or insulating condition on the field selects the fastest-growing mode of the TG dynamo,¹³ the magnetic boundary conditions shift the threshold of von Kármán flows,¹⁴ and soft-iron impellers select the mode of the von Kármán sodium dynamo,^{15,16} whose flow the TG vortex models.⁴ Both literatures ask how walls shape a flow or a field. Neither follows the share of the dissipation that walls take from a turbulent flow with a dynamical field as the Reynolds number grows.

Whether that share survives as the viscosity vanishes is a question about the walls themselves. For flows driven by moving walls, rigorous upper bounds on the dissipation do not depend on the viscosity at high Reynolds number;¹⁷ they bound the total, not where it is made. Kato showed that a Navier–Stokes flow in a bounded domain converges to a smooth Euler solution if and only if the dissipation in a layer of thickness proportional to the viscosity vanishes.¹⁸ In two-dimensional hydrodynamics, Nguyen van yen, Farge and Schneider found dissipating structures produced at no-slip walls whose dissipation appears to tend to a finite value as Re grows,¹⁹ and later work traced it to a small-scale instability of the boundary layer that drives a wall vorticity flux of order Re .²⁰ Confined MHD turbulence has been simulated with volume penalisation in two²¹ and three²² dimensions. We are not aware of a comparison that changes only the wall condition and follows the answer as Re increases.

We therefore ask: in decaying three-dimensional MHD turbulence, does a no-slip wall take a share of the dissipation that holds as Re grows (“walls”), or does that share fall, so that the periodic picture of a bulk cascade survives confinement (“cascade”)?

The answer rests on a construction that isolates the walls. The Taylor–Green (TG) MHD flows of Lee *et al.*³ and Pouquet *et al.*,⁴ which generalize the hydrodynamic TG vortex,²³ are symmetric about the faces of $[0, \pi]^3$. On those faces the velocity is impermeable and free-slipping, and the magnetic field has the parity of a perfect conductor or, for a second field, of a pseudo-vacuum. The periodic flow restricted to $[0, \pi]^3$ is therefore exactly the flow in a closed box with those walls. We solve that box

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directly, with walls on lattice nodes, and check that it reproduces the periodic flow to round-off. We then change the velocity condition from free slip to no slip and nothing else. The difference between the two boxes is the effect of the walls.

A question with two outcomes invites an analysis flexible enough to find either. We therefore fixed the observable, the resolution criterion and the decision rule in version-controlled job scripts before the runs they govern,^{24,25} and we report them as fixed. Section II describes the problem, the method and the protocol, Sec. III the verification, Sec. IV the results, and Sec. V what they settle and what they do not. In brief, the no-slip excess is robust: it is positive at every Re and under every definition we tried, it is viscous, and it comes with less near-wall Ohmic dissipation than free slip, which a second pre-registered test traces to the stretching of the field that the no-slip wall removes. The pre-registered verdict is “cascade”. But that verdict rests on a switch between two maxima of the dissipation, and over the factor of eight in Re that one GPU affords, the trend of the wall share depends on when it is measured.

II. PROBLEM, METHOD AND PROTOCOL

A. The flow and the box

We solve the incompressible visco-resistive MHD equations in units where the initial velocity amplitude and the density are one,

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + (\nabla \times \mathbf{B}) \times \mathbf{B} + \nu \nabla^2 \mathbf{v}, \quad (1)$$

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \quad (2)$$

with $\nabla \cdot \mathbf{v} = \nabla \cdot \mathbf{B} = 0$ and $\nu = \eta = 1/\text{Re}$, so the magnetic Prandtl number is one and $\text{Re} = 1000$ is run C2 of Ref. 4. Lengths are in units of $1/k_0$, where $k_0 = 1$ is the wavenumber of the TG velocity below, so the box has side π ; velocities are in units of its initial amplitude v_0 ; times are in units of $1/(k_0 v_0)$, about one large-eddy turnover time; and the magnetic field is in Alfvén units, $\mathbf{B}/\sqrt{\mu_0 \rho}$, which is why Eq. (1) carries no μ_0 . Every quantity is therefore dimensionless: $\text{Re} = v_0/(k_0 \nu)$, energies are per unit mass, and dissipation rates are in units of $k_0 v_0^3$. Most of what follows needs no units at all, being shares of the total dissipation or densities relative to the box mean. The initial conditions are those of Refs. 3 and 4:

$$\mathbf{v} = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0), \quad (3)$$

$$\mathbf{B}_C = b_0 (\sin 2x \cos 2y \cos 2z, \cos 2x \sin 2y \cos 2z, -2 \cos 2x \cos 2y \sin 2z), \quad (4)$$

$$\mathbf{B}_I = b_0 (\cos x \sin y \sin z, \sin x \cos y \sin z, -2 \sin x \sin y \cos z), \quad (5)$$

with $b_0 = 1/\sqrt{3}$, so that the kinetic and magnetic energies, $E_V = \langle |\mathbf{v}|^2 \rangle / 2$ and $E_M = \langle |\mathbf{B}|^2 \rangle / 2$, are both $1/8$ at $t = 0$.

TABLE I. Runs: two grids per rung. δ/h is the number of lattice spacings in the layer thickness $\delta = \text{Re}^{-1/2}$; $\tau - \frac{1}{2}$ is the fluid relaxation parameter; steps to $t = 10$ and the largest lattice Mach number over A and B are for the finer grid; “bits” is the floating-point precision.

Re	N	δ/h	$\tau - \frac{1}{2}$	steps	Ma _{max}	bits
125	128, 192	3.62, 5.44	0.0485, 0.0730	12159	0.094	64
250	192, 256	3.85, 5.13	0.0365, 0.0487	16234	0.102	64
500	256, 384	3.63, 5.45	0.0244, 0.0366	24383	0.109	64
1000	384, 512	3.86, 5.14	0.0183, 0.0244	32531	0.117	64
2000	512, 640	3.64, 4.55	0.0122, 0.0153	40680	0.133	32

On the six faces of $[0, \pi]^3$ the velocity (3) has $\mathbf{v} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{v}_t = 0$: an impermeable, free-slip wall. The field $\mathbf{B}_C$ has $\mathbf{B} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{B}_t = 0$. This is what a perfectly conducting wall imposes, since $\mathbf{n} \times \mathbf{E} = 0$ with Ohm’s law $\mathbf{E} = \eta \mathbf{j} - \mathbf{v} \times \mathbf{B}$ and $\mathbf{v} \cdot \mathbf{n} = \mathbf{B} \cdot \mathbf{n} = 0$ gives $\mathbf{n} \times \mathbf{j} = 0$, and it holds with either velocity condition. The field $\mathbf{B}_I$ has $\mathbf{B} \times \mathbf{n} = 0$ and $\partial_n B_n = 0$, the pseudo-vacuum condition. These parities are preserved by Eqs. (1)–(2). The boxes are:

A: free slip, conducting walls, $\mathbf{B}_C$: an exact restriction of the periodic flow, and the control;

B: no slip, conducting walls, $\mathbf{B}_C$: the experiment, which differs from A in the velocity condition only;

C: no slip, no field: the hydrodynamic reference;

A', B': as A and B, with $\mathbf{B}_I$ and pseudo-vacuum walls.

The TG velocity does not vanish on the faces, so a no-slip box starts impulsively: its wall nodes are set at rest at $t = 0$, and a vortex sheet on every wall dominates the early dissipation. In B the dissipation rate at $t = 0$ is $\varepsilon = 0.30, 0.22, 0.15$ and 0.09 at $\text{Re} = 250, 500, 1000$ and 2000 , 95–98% of it within $\text{Re}^{-1/2}$ of the walls.

B. Lattice Boltzmann discretization

The velocity is carried by a D3Q27 distribution with a central-moment collision in a shifted basis,^{26,27} and the Maxwell stress enters its equilibrium.²⁸ The field is carried by Dellar’s vector-valued distribution²⁸ on D3Q7 with a single-relaxation-time collision; ν and η set the two relaxation times.²⁹ The box has N nodes a side with the walls on nodes 0 and $N - 1$, so the spacing is $h = \pi/(N - 1)$. The lattice velocity of the unit velocity is $u_0 = 0.05$, one step is $\Delta t = u_0 h$, and $\nu_{\text{lat}} = u_0/(h \text{Re})$; the fluid relaxation parameter is $\tau - \frac{1}{2} = 3\nu_{\text{lat}}$, and the magnetic one is $4\eta_{\text{lat}}$ on D3Q7. Table I lists the runs.

All walls are on nodes, and each wall node collides like any other once its unknown populations are set. For free slip these are the mirror images of their partners about the wall (specular reflection). For no slip we use the

regularized velocity condition of Latt *et al.*³⁰ with zero wall velocity; at the twelve edges and eight corners, where it has no density stencil, the density is set to one. For the field we use a parity wall. Write the field component a as carried by its own populations $g_i^{(a)}$. At a node on a wall normal to axis k , every population i entering the box is replaced by

$$g_i^{(a)} \leftarrow s_{ak} g_{i^*}^{(a)}, \quad (6)$$

where i^* is the mirror of i about the wall and $s_{ak} = -1$ if the wall flips component a (the normal one, $a = k$, at a conducting wall; the tangential ones, $a \neq k$, at a pseudo-vacuum wall) and $+1$ otherwise. At edges and corners the mirrors compose and the signs multiply. A conducting wall thus flips only the normal component, and a pseudo-vacuum wall only the tangential ones. Delar has given moment-based magnetic boundary conditions for the same distribution;³¹ Eq. (6) is a population-level mirror that imposes a parity on any combination of faces. With free slip and either parity, the box of N nodes reproduces the periodic 2π box of $2(N-1)$ nodes to 10^{-14} , node for node and transient included (supplementary material, Sec. S1, gives this and two further tests).

The method exists in two implementations that share no source code: one in C++20 on Kokkos,^{32,33} one in CUDA. Their host builds agree to every printed digit, and the CUDA build on the device agrees with the Kokkos reference to within 5.5×10^{-16} of each diagnostic's scale. The production runs used NVIDIA A100 (80 GB) GPUs: double precision up to $\text{Re} = 1000$, and single precision at $\text{Re} = 2000$, where $N = 640$ fits in memory only in single precision. They took 34 GPU-hours.

C. Diagnostics

The dissipation rate is $\varepsilon = \nu \langle |\omega|^2 \rangle + \eta \langle |\mathbf{j}|^2 \rangle$, with $\omega = \nabla \times \mathbf{v}$ and $\mathbf{j} = \nabla \times \mathbf{B}$. The box average $\langle \cdot \rangle$ uses trapezoidal weights, $\frac{1}{2}$ on a face, $\frac{1}{4}$ on an edge and $\frac{1}{8}$ at a corner, which makes the box average of a mirror-symmetric field equal to its periodic average exactly. Derivatives are second-order central differences, extended at a wall by the wall's own rule: the mirror for a free-slip velocity and for the magnetic parity, and a one-sided second-order stencil for a no-slip velocity.

Each node is at distance $d = kh$ from its nearest wall, $k = 0, 1, \dots$. With the trapezoidal weights, the nodes with $d = kh$ fill exactly the part of the box at distances between $(k - \frac{1}{2})h$ and $(k + \frac{1}{2})h$ from the walls (between 0 and $h/2$ for the wall nodes). If e_k is the dissipation in layer k , the dissipation within a distance rh of the walls is

$$\mathcal{E}(r) = \sum_{j < k} e_j + (r - k + \frac{1}{2}) e_k, \quad k = \lfloor r + \frac{1}{2} \rfloor, \quad (7)$$

that is, linear within each layer (with $\mathcal{E} = 2re_0$ for $r < \frac{1}{2}$). The share of the dissipation within $m\delta$ of the walls,

$\delta = \text{Re}^{-1/2}$, is

$$f_m(t) = \mathcal{E}(m\delta/h) / \sum_k e_k, \quad m = 1, 2, 4, \quad (8)$$

and it divides into a viscous and an Ohmic part, $f_m = f_m^\nu + f_m^\eta$, each a share of the total ε . The dissipation density relative to the box mean is a layer's share of ε over its share of the volume.

Counting whole node layers instead, every node with $kh < m\delta$, measures a layer of thickness $(\lceil m\delta/h \rceil - \frac{1}{2})h$ rather than $m\delta$. Our first ladder did, and failed its resolution gate at every rung for that reason alone (Sec. S3).

We also use, after the fact (Sec. IV E), the time-integrated share over a window $[t_0, t_1]$,

$$F_m = \int_{t_0}^{t_1} f_m \varepsilon dt / \int_{t_0}^{t_1} \varepsilon dt, \quad (9)$$

the fraction of all the energy dissipated in the window that was dissipated within $m\delta$ of the walls. It is evaluated with the trapezoidal rule over the probes, which are 0.05 or 0.1 apart, with every integrand interpolated to the window's edges.

The drivers also write the energy budgets layer by layer. With $e_M = |\mathbf{B}|^2/2$,

$$\partial_t e_M = S + T_M - \eta |\mathbf{j}|^2, \quad S = B_i B_j \partial_j v_i, \quad (10)$$

where S is the stretching of the field by the flow, the rate at which the field gains energy from it, and T_M collects the advection of e_M , its resistive transport and the compressible part of the induction. The transports integrate to zero over the box but not over a layer. The kinetic budget carries the Lorentz work $\mathbf{v} \cdot (\mathbf{j} \times \mathbf{B})$, which balances S over the box, beside pressure work, advection, viscous transport and the viscous dissipation. Within $m\delta$ of the walls each term is read as f_m is and integrated over a window. Products of central differences do not telescope, so the explicit transports close a layer's budget only to a residual: within δ , on the finer grids, 2.9% of its Ohmic dissipation or less (Sec. S7).

D. The pre-registered protocol

The two outcomes of Sec. I were stated, qualitatively, on 25 September 2026, before any production run. The following rules were committed with the job scripts before the runs they govern: the observable and the gate on 27 September; the estimator of f_m , corrected after the first ladder failed the gate for the reason given above, on 28 September, with the observable it estimates and the gate unchanged; the Mach control on 29 September; and the decision rule and the precision control for the top rung on 30 September, after the lower rungs had been run. GitHub's record of when each commit reached it is kept with the data (Sec. S2).

1. *Observable*: $B-A$ in f_1 at each run's turbulent dissipation peak. The peak is the maximum of ε after ε has first risen 5% above its running minimum, refined by a parabola through the three probes around the discrete maximum; f_m is interpolated to that time. The rule skips the impulsive start of a no-slip box; for free slip the peak is the global maximum. A run in which ε never rises 5% above its running minimum has no turbulent peak and no f_m .
2. *Resolution gate*: at every (box, Re), f_1 , f_2 and f_4 at the peak agree within 5% (relative) between the two grids.
3. *Band*: the grid differences of f_1 for A and for B, added.
4. *Decision*: “walls hold” if $B-A$ at the top rung lies within the combined bands of its value at the rung below; otherwise “cascade”.
5. *Controls*, each changing one thing: u_0 halved at Re = 500 and 1000, which halves the Mach number and $\tau - \frac{1}{2}$ together; and double against single precision at Re = 2000, $N = 512$. A control passes if it moves $B-A$ by less than the band.

The time-integrated share (9) was not part of the protocol; we introduce it in Sec. IV E because of what Sec. IV D found. A second test, of what causes the near-wall Ohmic deficit, was fixed in the same way on 2 October 2026, with the job script that ran it, and is stated where it is applied (Sec. IV G); so were four controls, on 3 October (Sec. IV H).

III. VERIFICATION

The free-slip box A at Re = 1000 is run C2 of Pouquet *et al.*⁴ Its minimum ratio of magnetic to kinetic energy is $E_M/E_V = 0.3695$ and 0.3693 at $N = 384$ and 512 , reached at $t = 7.3$, against their 0.35 . The first maximum of Ω_M/Ω_V after its initial fall is 2.154 and 2.159 , against their “2.” Here Ω_V and Ω_M are the kinetic and magnetic enstrophies. Our ratio is converged to four figures between the two grids and lies 5.5% above theirs, which is quoted to two figures. We attribute the gap to the resolution of C2, their coarsest run: its own resolution measure, $\delta k_{\max} = 0.8$, falls below the $\delta k_{\max} \geq 2$ by which they call their finer runs well resolved, and our coarser grids move the same way, to 0.357 at $N = 97$ and 0.329 at $N = 65$ (Sec. S1).

Every rung of A and B with a turbulent peak passes the resolution gate, the worst difference being 3.0% in f_1 of A at Re = 2000 (Table S2). Without a turbulent peak by $t = 10$ on either grid are B at Re = 125 and C at Re ≤ 500 ; C at Re = 2000 is still rising at $t = 10$. Halving u_0 moves $B-A$ by -0.0014 at Re = 500 and by -2×10^{-5} at Re = 1000, against bands of 0.0030 and

0.0004 . Double precision at Re = 2000 moves it by less than 10^{-5} against 0.0021 , although the two runs differ by up to 4×10^{-5} (relative) in their time series. All three controls pass. Over every run, the total mass drifts by at most 4.4×10^{-5} (relative), and the r.m.s. of $\nabla \cdot \mathbf{B}$ stays below 1% of the r.m.s. of $|\mathbf{j}|$. The largest lattice Mach number is 0.133 , in A at Re = 2000 (Table I).

IV. RESULTS

A. Where the dissipation sits

Figure 1(a) shows how the dissipation is spread across the box at Re = 1000, integrated over $t = 2-10$. At the walls, the dissipation density is 8.50 times the box mean in C, 2.43 times in B and 0.38 times in A. The free-slip conducting walls are quieter than the bulk. The no-slip walls of the MHD box are more active than the bulk, but much less so than those of the hydrodynamic box. In C the density falls below the mean beyond $d \approx 4\delta$: in that box the walls do most of the dissipating. Over $t = 2-10$, C dissipates 33% of its energy within δ of the walls and B 9.9%. At its own turbulent peak ($t = 3.20$), C's f_1 is 0.50 . C carries half of B's initial energy, since it has no field. This is the one rung at which both boxes have a turbulent peak, so it is the one rung at which they can be compared.

Figure 1(b) follows the density at the walls with Re. In B it rises monotonically, 1.44, 1.82, 2.43 and 3.13 times the mean at Re = 250–2000. In A it stays near 0.38 and falls to 0.34 at Re = 2000. B's wall density rises monotonically in each of the five time windows of Sec. IV E, as $\text{Re}^{0.25}$ to $\text{Re}^{0.44}$ depending on the window. A's changes as $\text{Re}^{-0.02}$ to $\text{Re}^{-0.11}$.

B. The no-slip excess and what it is made of

Table II and Fig. 2(b) give the time-integrated share F_1 over $t = 2-10$. No-slip walls take more of the dissipation next to them at every rung: $B-A = 0.059, 0.055, 0.054$ and 0.048 at Re = 250–2000, with grid bands of at most 0.0009 . The excess is viscous. Its viscous part is $+0.072$ to $+0.059$ and its Ohmic part is *negative*, -0.014 to -0.011 . That deficit is not an artefact of dividing by a larger total. Over the window the no-slip box dissipates *less* energy than the free-slip one ($0.55-0.72$ of the initial energy, against $0.62-0.75$), so its near-wall Ohmic dissipation is lower in absolute terms too, $0.55-0.59$ of A's. In A the Ohmic part is the larger of the two near the walls: 0.039 against 0.030 at Re = 250, and 0.026 against 0.012 at Re = 2000.

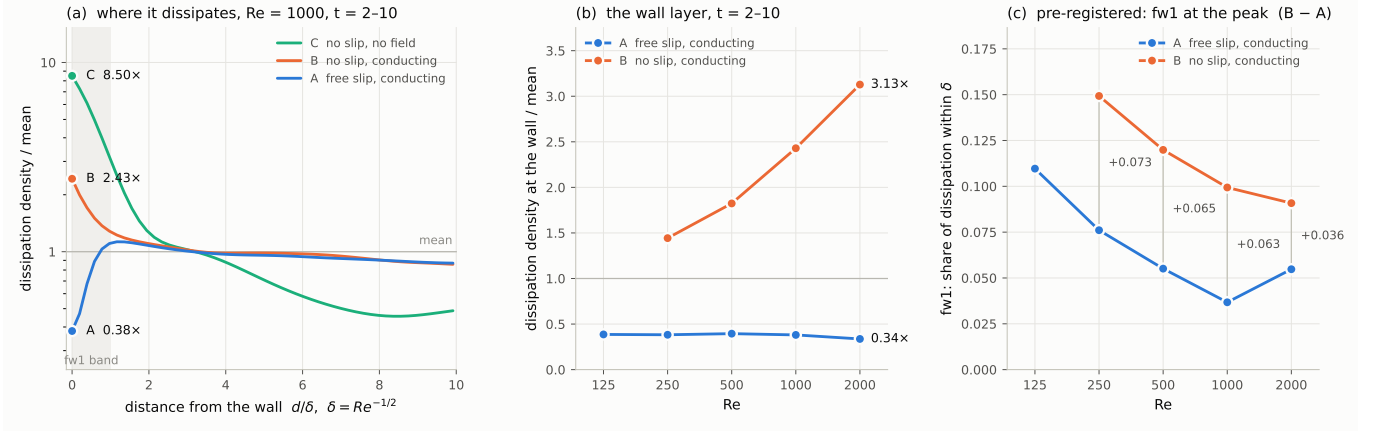


FIG. 1. Where the dissipation sits. (a) Dissipation density relative to the box mean against distance from the walls, d/δ with $\delta = Re^{-1/2}$, at $Re = 1000$, integrated over $t = 2-10$; the shaded layer is that of f_1 . (b) The density at the walls against Re , over the same window. (c) The pre-registered observable: f_1 at each run's turbulent dissipation peak, with B-A at each rung. Each point is from the finer grid of its rung (Table I).

TABLE II. Results on the finer grid of each rung. At the peak: t_p , the time of the turbulent dissipation peak in units of $1/(k_0 v_0)$, and B-A in f_1 there, the pre-registered observable. Integrated: the share F_1 of the dissipation over $t = 2-10$ within δ of the walls, and the viscous and Ohmic parts of B-A. The grid bands of the integrated B-A are at most 0.0009.

Re	at the peak (pre-registered)			integrated, $t = 2-10$				
	t_p^A	t_p^B	B-A	F^A	F^B	B-A	viscous	Ohmic
250	2.01	1.90	+0.073	0.069	0.128	+0.059	+0.072	-0.014
500	2.23	2.14	+0.065	0.056	0.111	+0.055	+0.069	-0.014
1000	4.06	2.34	+0.063	0.046	0.099	+0.054	+0.065	-0.012
2000	4.45	4.53	+0.036	0.038	0.086	+0.048	+0.059	-0.011

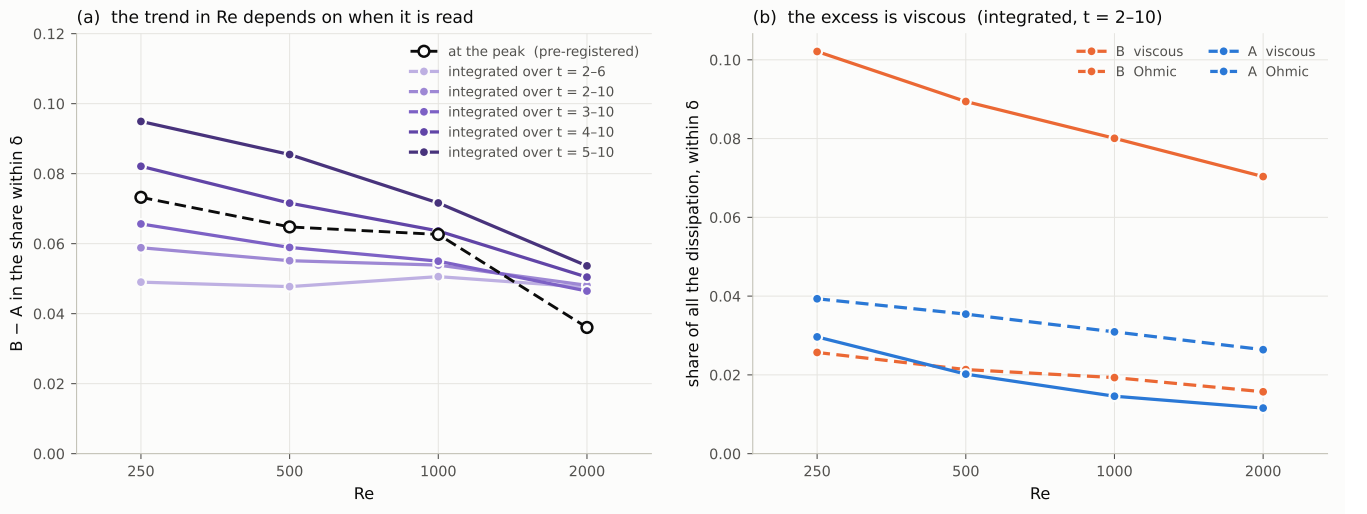


FIG. 2. (a) B-A in the share of the dissipation within δ of the walls against Re : at the turbulent peak, as pre-registered (dashed), and integrated over five time windows, after the fact (light to dark: later windows). (b) The integrated share F_1 over $t = 2-10$ for A and B, divided into its viscous and Ohmic parts. Grid bands are at most 0.001, smaller than the symbols.

C. The pre-registered verdict

Read at the turbulent peak, as registered, $B-A$ is 0.0732, 0.0648, 0.0626 and 0.0361 at $Re = 250-2000$ [Fig. 1(c)], with bands of 0.0020, 0.0030, 0.0004 and 0.0021. From $Re = 1000$ to 2000 it falls by 0.0266, 10.5 times the two rungs' bands added. By the rule fixed before the $Re = 2000$ runs, the outcome is "cascade".

D. The peak switches

Figure 3 shows why the last step is not what it seems. The dissipation of A has an early maximum near $t = 2.2$ and a later one at $t = 3.6-4.4$. At $Re = 500$ they are 5% apart; from $Re = 1000$ the later one is the larger, and A 's peak time jumps from 2.23 to 4.06. B 's peak moves the same way between $Re = 1000$ and 2000, from 2.34 to 4.53: at $Re = 2000$ its early rise is only a shoulder [Fig. 3(b)].

And f_1 does not vary slowly through either maximum. A 's oscillates with a period near 1.3, between 0.022 and 0.060 at $Re = 2000$. B 's falls from 0.32 at $t = 1.5$ to 0.06 at $t = 3.7$ as the start-up layers decay, then rises to about 0.11 [Fig. 3(c, d)]. The pre-registered reading therefore compares the two boxes at different times: at $Re = 1000$, B at $t = 2.34$ and A at $t = 4.06$. Between $Re = 1000$ and 2000 it moved from a time at which $B-A$, compared at equal times, is near 0.06 at both Reynolds numbers (0.059 and 0.064 at $t = 2.3$) to one at which it is near 0.03 at both (0.035 and 0.028 at $t = 4.5$). At equal times, $B-A$ does not fall with Re up to $t \approx 4$: at $t = 3.5$ it is 0.034–0.036 at every rung, and at $t = 2$ it rises, from 0.062 to 0.097. It falls only from $t \approx 4.5$ on: at $t = 8$, from 0.107 to 0.046.

None of the checks of Sec. III could see this. The gate and the Mach and precision controls each compare runs at one Re , and those switch together: A 's peak is at $t = 4.05$ and 4.06 on the two grids at $Re = 1000$, and at 4.04 with u_0 halved.

E. The trend depends on when it is read

The time-integrated share (9) has no maximum to pick, but it needs a window. Table III and Fig. 2(a) give $B-A$ over five. Over $t = 2-6$, around the turbulent peaks ($t_p = 1.90-4.53$), $B-A$ is flat: 0.049, 0.048, 0.051 and 0.048, within 6% at every rung. Windows reaching further into the decay give a falling trend, -18% over $t = 2-10$ and -43% over $t = 5-10$ from $Re = 250$ to 2000.

A fixed window compares different phases of the decay, since the evolution slows as Re grows: A 's peak moves from $t = 2.0$ to 4.45. Aligning each run's window on its own energy, from the time when the total energy E_T has fallen to a fraction f_a of its initial value to the time when it has fallen to f_b , does not settle the question either. From $E_T/E_T(0) = 0.9$ to 0.5, $B-A$ falls at every step,

from 0.166 to 0.048. From 0.6 to 0.4 it falls and then rises: 0.048, 0.029, 0.034 and 0.053. Applied to the default window $t = 2-10$, the registered rule also gives "cascade", with $B-A$ falling from $Re = 1000$ to 2000 by 0.0058 against bands of 0.0017. But the fall is 11% rather than 42%, and over $t = 2-6$ there is no fall at all.

The thickness of the layer does not change this. Within 2δ and 4δ of the walls, $B-A$ is positive in every reading at every rung, and its trend again depends on when it is read: from $Re = 250$ to 2000 it falls by 67–68% at the peak and by 12–42% over the windows that reach into the decay, and changes by $+2\%$ and -20% over $t = 2-6$ (Table S3).

F. What the fields show

The shares above are integrals of local fields. Figure 4 shows the fields themselves at $Re = 1000$ and $t = 2.3$, B 's turbulent peak, and Sec. S9 repeats it at $t = 4.6$, by the later maxima. They come from reruns of the production runs that also write the full fields, and the reruns reproduce the production time series exactly, at every one of the 48 probes to $t = 4.7$. The local viscous and Ohmic dissipation is computed from the fields with the drivers' own derivatives at the walls; summed over the box, it reproduces ε to 2×10^{-8} (checked at $N = 65$ for all three boxes).

In the bulk the two MHD boxes are alike. In the section $y = \pi/4$ they show the same current and vorticity sheets, and their plane means differ by less than 15%: viscous 0.37 and 0.39 of the box mean in A and B at $t = 2.3$, Ohmic 0.49 and 0.42. They differ next to the walls. In the plane 0.58δ above the bottom wall, B 's viscous dissipation is broad and strong, 1.49 of its box mean against A 's 0.53, and its Ohmic dissipation is weaker, 0.39 against 0.61. In absolute terms, B 's viscous dissipation there is 3.5 times A 's and its Ohmic 0.79 of A 's. In A the plane is crossed by current sheets, and bright ones run along its edges, where it meets the vertical walls. At $t = 4.6$ the contrast remains: B 's viscous dissipation there is 6.8 times A 's and its Ohmic 0.68 of A 's. These are single planes at two instants, but they point the same way as the time-integrated profiles of Sec. IV B, in which B 's near-wall Ohmic dissipation is 0.55–0.59 of A 's.

The box without a field is different in kind. Its sections glow at all four walls while its bulk is nearly uniform, and its near-wall plane dissipates at 6.7 times its box mean at $t = 2.3$ and 3.3 times at $t = 4.6$, about 3.5 and 3.2 times B 's relative level.

By the symmetries of the TG flow the six walls fall into two classes, the two horizontal walls and the four vertical ones. The near-wall planes show the bottom wall, of the first class; the sections cross both.

Figure 5 shows the same fields in three dimensions. A plane shows a sheet only where it cuts it; these views project the whole box, each pixel showing the largest local dissipation along its line of sight. They are computed

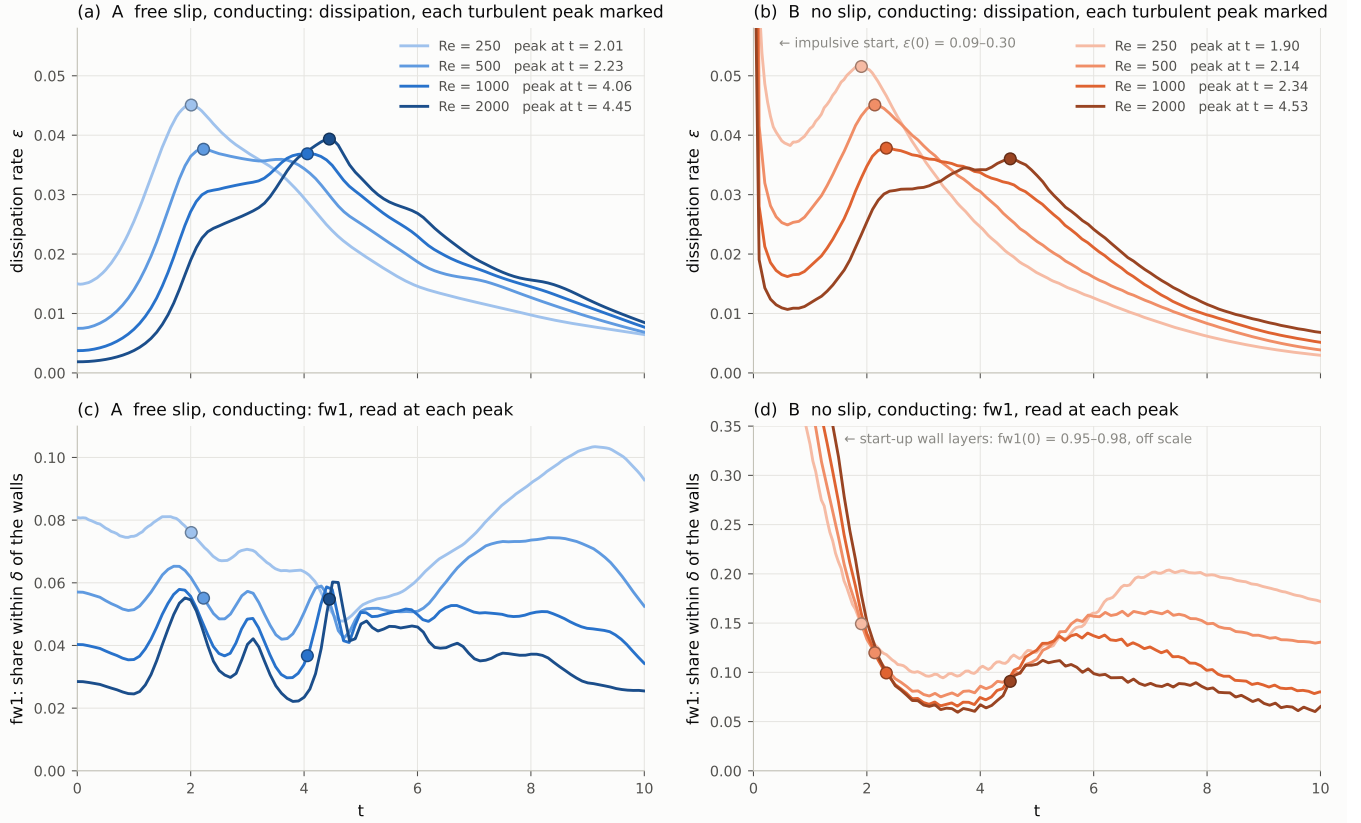


FIG. 3. The turbulent peak switches. The dissipation rate $\epsilon(t)$, in units of $k_0 v_0^3$ (a, b), and the share within δ of the walls, $f_1(t)$ (c, d), for the free-slip box A (a, c) and the no-slip box B (b, d) at $Re = 250-2000$ (light to dark). Dots mark each run's turbulent peak and the f_1 read there. A's early maximum is the larger up to $Re = 500$ and its later one from $Re = 1000$; B switches between $Re = 1000$ and 2000 . B starts off scale at $t = 0$ (impulsive start). Time is in units of $1/(k_0 v_0)$.

TABLE III. B–A in the share within δ of the walls: at the turbulent peak, integrated over fixed time windows, and integrated over windows bounded by each run's own total energy. The last column is the change from $Re = 250$ to 2000 . Grid bands are at most 0.0010 for the time windows and 0.0027 for the energy windows.

B–A in the share within δ	250	500	1000	2000	change (%)
at the peak (pre-registered)	0.073	0.065	0.063	0.036	–51
integrated, $t = 2-6$	0.049	0.048	0.051	0.048	–3
integrated, $t = 2-10$	0.059	0.055	0.054	0.048	–18
integrated, $t = 3-10$	0.066	0.059	0.055	0.046	–29
integrated, $t = 4-10$	0.082	0.072	0.064	0.050	–39
integrated, $t = 5-10$	0.095	0.085	0.072	0.054	–43
integrated, $E/E_0 = 0.9-0.5$	0.166	0.113	0.071	0.048	–71
integrated, $E/E_0 = 0.6-0.4$	0.048	0.029	0.033	0.053	+11

from the dumps with the same derivatives as the planes, and every plane of the snapshots lies inside its volume exactly. A projection of the maximum shows where the sheets are, not how much they dissipate; the shares are those of Sec. IV B.

The views confirm the planes and add to them. The current sheets fill the bulk of both MHD boxes alike, at both times, and by $t = 4.6$ both have broken up into finer

sheets (Fig. S2). The viscous dissipation of the no-slip box lines its walls, all of them and not only the bottom one of the planes, and by $t = 4.6$ it forms ring-like patterns on its faces; the free-slip box has no such lining. The box without a field is dominated by its walls and edges, as its sections were.

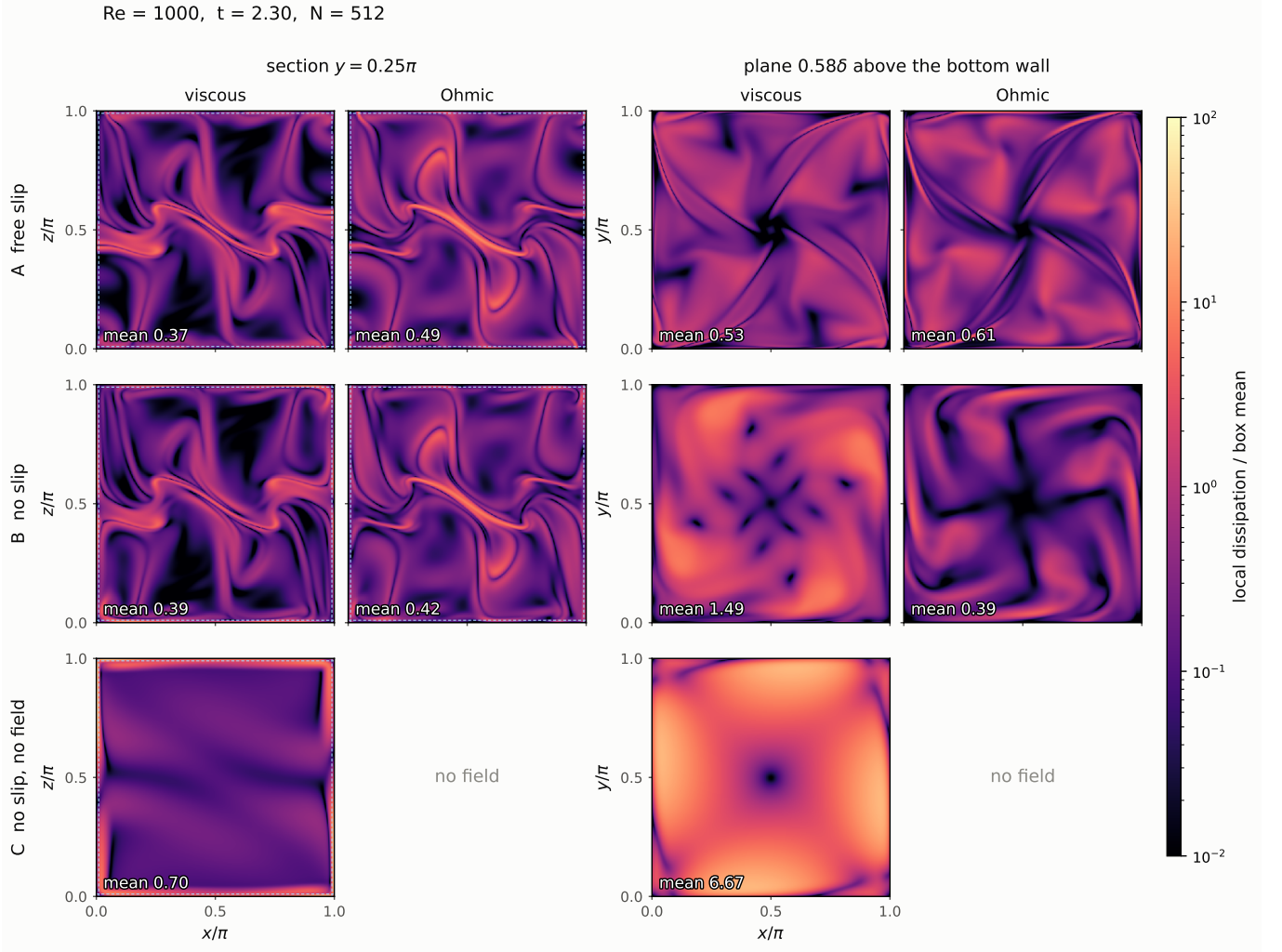


FIG. 4. Where the dissipation sits, at $Re = 1000$ and $t = 2.3$ (B’s turbulent peak), on the finer grid ($N = 512$). Rows: the free-slip box A, the no-slip box B and the box without a field C. Columns: the viscous and Ohmic dissipation, $\nu|\omega|^2$ and $\eta|j|^2$, in the vertical section $y = \pi/4$, with dashed lines at δ from its four walls, and in the plane 0.58δ above the bottom wall. One logarithmic colour scale, relative to each box’s mean dissipation at that time; each panel gives its plane’s mean.

G. What feeds the near-wall Ohmic dissipation

The snapshots suggest that the current sheets next to a free-slip wall need the tangential flow along it, but they cannot test that. The budget (10) can, and we fixed the test before running it, on round 2’s sixteen A and B runs again: reruns that reproduce round 2’s time series exactly and also write the budget. Over $t = 2$ –10 and within δ of the walls, let Ω_X and S_X be the time-integrated Ohmic dissipation and stretching of box X. The share of the Ohmic deficit that the stretching deficit accounts for is

$$\phi = (S_A - S_B)/(\Omega_A - \Omega_B), \quad (11)$$

with a band from the two grids of each rung. The verdict was to be “stretching” if ϕ minus its band exceeds $\frac{1}{2}$ at every rung, “transport” if ϕ plus its band is below $\frac{1}{2}$ at every rung, and undecided otherwise. Transport is

defined as whatever closes the layer’s budget, so ϕ needs only S and the Ohmic dissipation, both products of first derivatives.

The verdict is “stretching”. At $Re = 250, 500, 1000$ and 2000 , $\phi = 1.86, 1.55, 1.47$ and 1.84 , with bands of at most 0.052 [Fig. 6(b), Table S4]. The stretching deficit exceeds the Ohmic deficit because the two boxes’ stretching differs in sign, not only in size. Within δ of a free-slip wall the flow feeds the field, S_A being 0.56 – 0.67 of Ω_A ; within δ of a no-slip wall S_B is negative, -0.08 to -0.40 of Ω_B , at every rung and on both grids, and the Lorentz work has the opposite signs. The field does work on the flow there. The rest of the deficit is balanced by the layer’s stored magnetic energy at low Re and by transport at high Re .

Figure 6(a) shows where, at $Re = 1000$. A’s stretching is largest on the wall, 0.39 of the mean dissipation, and

Re = 1000, $t = 2.30$, $N = 512$ (block maximum over 2^3)

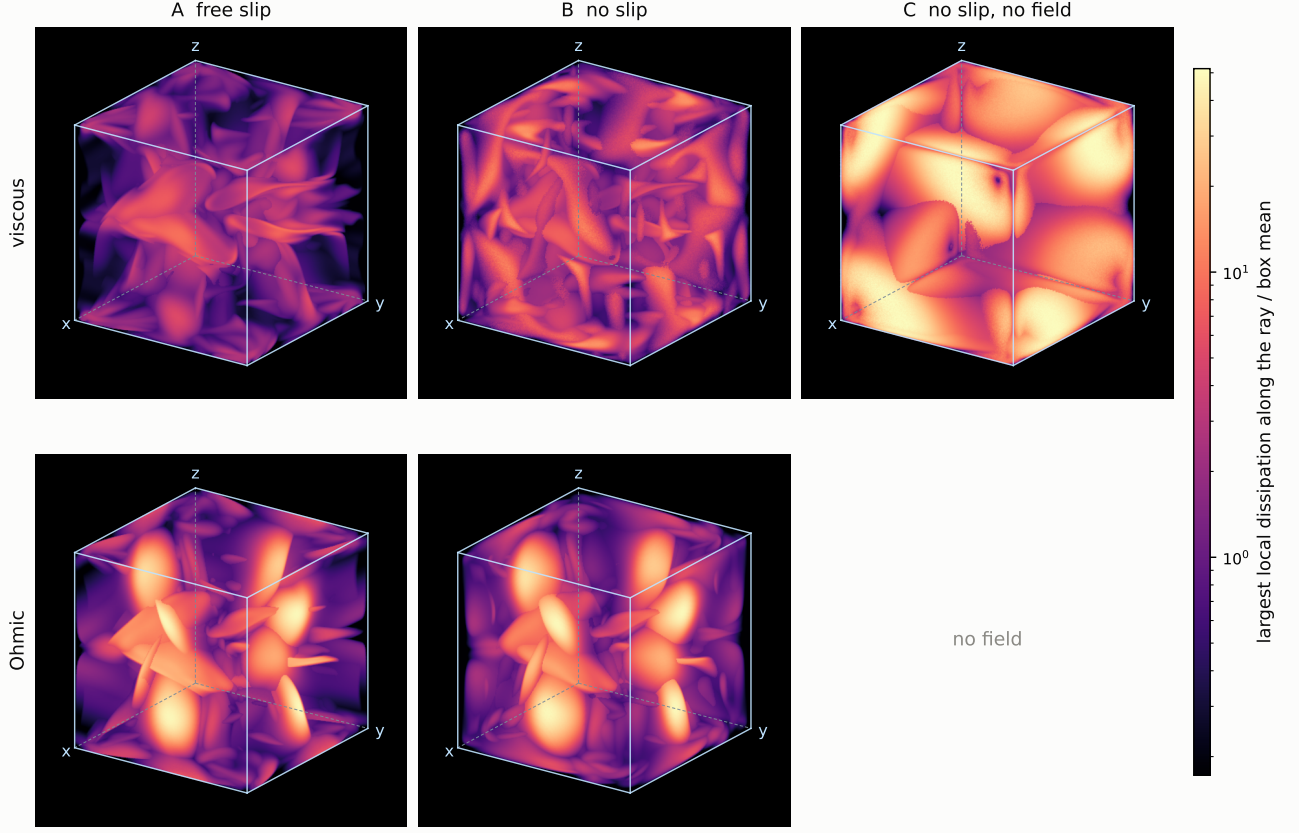


FIG. 5. Three-dimensional views of the local dissipation at $\text{Re} = 1000$ and $t = 2.3$, on the finer grid. Columns: the boxes A, B and C; rows: the viscous and Ohmic dissipation. Each pixel is the largest value along an oblique line of sight (maximum-intensity projection) through the volume, reduced to 256^3 by the maximum over 2^3 blocks, relative to the box mean. The far half is dimmed as a depth cue, which must not be read as an intensity; hidden edges of the box are dashed. One logarithmic colour scale, shared with Fig. S2.

falls to zero at 2.5δ . B's vanishes on the wall, where the velocity and its tangential derivatives vanish and the field is tangential; it is negative out to 0.8δ and positive beyond. The Ohmic dissipation peaks at 1.2δ in A and at 1.9δ in B: the no-slip wall moves it away from itself.

The kinetic budget of B's layer completes the picture [Fig. 6(c)]. Viscous transport from outside the layer supplies 54–66% of its viscous dissipation and the Lorentz work 14–36%, a share that rises with Re over this window; pressure work, advection and the layer's own kinetic energy supply the rest.

H. Four controls

Four controls, each with its rule fixed before its runs (Sec. S8), test what these results might owe to the way they were made.

The start. With the walls of B brought to rest over

0.2 or 1.0 time units instead of at once, B–A over $t = 2$ –10 changes by -0.2% and $+2.2\%$ at $\text{Re} = 2000$, and by $+5.0\%$ and $+20.3\%$ at $\text{Re} = 500$. The rule allowed 10%, so by it the start is a factor of the excess at the lower rung. A gentler start makes the excess larger, not smaller, and its effect falls with Re .

The field at matched energy. Started with B's initial energy, $E_V(0) = \frac{1}{4}$, the box without a field puts 2.3–2.6 times B's share of its dissipation within δ of the walls, at every rung (0.21–0.29 against 0.086–0.13), and it now has a turbulent peak at $\text{Re} = 500, 1000$ and 2000 . The field moves dissipation away from the walls at matched energy too.

The Mach number at the top rung. Lowering u_0 to 0.04 at $\text{Re} = 2000$, which takes the largest lattice Mach number from 0.133 to 0.107, moves B–A by 7×10^{-5} against a band of 9×10^{-4} , and ϕ by 0.003 against 0.052.

The magnetic Prandtl number. At $\text{Pm} = 0.5$ and 2 ($\text{Re} = 1000$), B–A is 0.060 and 0.052, against 0.054

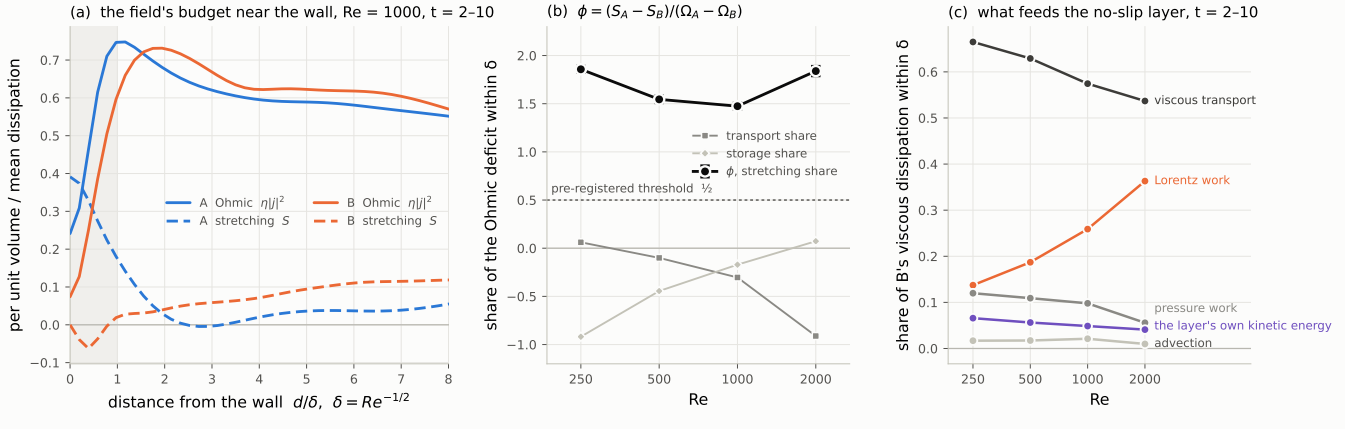


FIG. 6. What feeds the near-wall Ohmic dissipation, over $t = 2-10$, on the finer grid of each rung. (a) At $Re = 1000$, the stretching $S = B_i B_j \partial_j v_i$ (dashed) and the Ohmic dissipation $\eta|j|^2$ (solid) per unit volume, relative to the box-mean dissipation, against d/δ , for A and B; the shaded band is δ . (b) ϕ , the share of the near-wall Ohmic deficit that the stretching deficit accounts for [Eq. (11)], with its two-grid band, mostly smaller than the symbols, against the pre-registered threshold $\frac{1}{2}$; the transport and storage shares are its complements, the three adding to one. (c) What supplies B's viscous dissipation within δ , as shares of it.

at $Pm = 1$; it is viscous, with a negative Ohmic part, and ϕ is 2.01 and 1.22, against 1.47. The mechanism of Sec. IV G holds at both, the stretching deficit coming closer to the Ohmic deficit as Pm grows.

V. DISCUSSION

What the data settle. Every measure we tried agrees on four statements. First, a no-slip wall takes more of the dissipation next to it than a free-slip one. B–A is positive and far outside its band at every Re : at the peak, at every probe from $t = 1.5$ to 10 (where it is at least 0.019), over every time and energy window, and within 2δ and 4δ as within δ . Second, the excess is viscous, and the near-wall Ohmic dissipation is lower, not higher, than at a free-slip conducting wall, by 40–45% in absolute terms. Third, the no-slip wall layer intensifies with Re : B's wall dissipation density rises monotonically in every window, while A's stays near 0.4 of the mean. Fourth, at the one rung where the hydrodynamic box is turbulent, the field moves dissipation away from the walls: the box without a field dissipates 33% of its energy within δ of the walls, the box with one 9.9%. A fifth statement comes from the second test: the Ohmic deficit is the stretching that the no-slip wall removes (Sec. IV G). All five survive the controls of Sec. IV H: a smooth start, matched initial energy, a lower Mach number at the top rung, and $Pm = 0.5$ and 2; the start changes the size of the excess, by up to 20% at $Re = 500$, but not its sign.

What they do not settle. The question that motivated the work was whether the wall share holds as Re grows, and it has a registered answer: “cascade”. That answer rests on the peak switching from B's early maximum to its later one. A reading that does not depend on one

instant depends instead on a window. B–A is flat over the window around the peaks and falls by 18–43% over windows that reach further into the decay, and aligning the windows on each run's energy moves the ambiguity rather than removing it. Over a factor of eight in Re , the trend of a decaying flow's wall share cannot be separated from the dependence of that share on the phase of the decay. We therefore claim neither outcome. The cascade verdict stands as the result of the test we registered, and we record that equally reasonable readings would not have produced it.

Scalings. Two limits frame the exponents. A laminar Prandtl layer beside a bulk whose dissipation does not depend on Re keeps a dissipation density, relative to the mean, that does not grow with Re , so its share falls with its volume, as $Re^{-1/2}$ ($Re^{-0.49}$ for our box). For the share to hold as $Re \rightarrow \infty$, the density must rise as $Re^{1/2}$ instead. Over $t = 2-10$, B's share falls as $Re^{-0.19}$, and its density at the wall rises as $Re^{0.25}$ to $Re^{0.44}$ depending on the window: between the two limits, and a factor of eight in Re does not show which, if either, it approaches.

Kato's layer, of thickness $\nu = \delta^2$, is $0.32h$ at $Re = 250$ and $0.10h$ at $Re = 2000$, so these data cannot resolve it. They bound it from above, since the share within δ contains it; and since his theorem concerns convergence to a smooth Euler solution,¹⁸ which a turbulent TG flow is not known to have, we use his layer as a scale and not his theorem as a test. MHD has a layer of that thickness of its own. Where a field crosses a no-slip wall, a Hartmann layer,^{8,9} $\sqrt{\nu\eta}/|B_n|$ thick, or $1/(Re|B_n|)$ at $Pm = 1$, dissipates per unit area an amount that does not vanish with the viscosity at fixed Pm (Sec. S6). Our conducting walls, with $\mathbf{B} \cdot \mathbf{n} = 0$, carry none; those of B' would, $0.14h$ thick at $Re = 1000$, and we have not resolved them. Whether a finite part of the dissipation remains at the

walls of A and B as $\text{Re} \rightarrow \infty$ cannot be decided from these data.

The mechanism of the Ohmic deficit. At a free-slip conducting wall the tangential flow reaches the wall, and with $\mathbf{B} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{B}_t = 0$ it stretches tangential field along it: the stretching is largest on the wall itself, and the snapshots show current sheets crossing the plane 0.58δ above it. A no-slip wall stops that flow. The test of Sec. IV G confirms that this is what cuts the near-wall Ohmic dissipation, and sharpens it: the no-slip wall does not only remove the production of magnetic energy next to it, it reverses it, so that within 0.8δ the field does work on the flow, and the layer dissipates that work viscously with the energy viscous transport brings in. Part of the viscous excess is that exchange: the Lorentz work supplies 14–36% of B’s near-wall viscous dissipation, and viscous transport most of the rest. What the budget does not settle is why the Lorentz work’s share of B’s layer grows with Re: that is a trend read through one window, of the kind Sec. IV E warns against.

What the protocol bought. Had the decision rule not been fixed in advance, any window of Table III could have been chosen after seeing the data. Over $t = 2$ –6 it would have supported “walls hold”, over $t = 5$ –10 “cascade”. Fixing the rule did not make the observable robust, and it was not. But it made the fragility something to report rather than something to choose around.²⁵ In a decaying flow a quantity read “at the peak” compares runs at different times whenever the dissipation has more than one maximum of similar height, and TG flows, with their sequences of sheet formation and collision,^{3,4} are such flows. Before reading a trend from one instant of a transient, the time series at every parameter value should be inspected, and more than one definition of “when” should be tried.

Limitations. One initial condition, of high symmetry, in one realization; $\text{Pm} = 1$ on the ladder, with checks at 0.5 and 2 at one rung only; and $\text{Re} \leq 2000$, set by the memory of one GPU, since resolving δ with about 4.5 spacings makes N grow as $\text{Re}^{1/2}$. The method is weakly compressible. The Mach controls bound its effect, but at $\text{Re} = 2000$ only for a 20% lower Mach number, which is as far as the wall’s stability floor allows on one GPU. The regularized edges and corners do not conserve mass exactly, though the drift stays below 4.4×10^{-5} . Only conducting walls are tested: the insulating pair A’, B’ cannot show whether the result needs a conducting wall, since B’ has not become turbulent by $t = 10$ (Sec. S6). A forced, statistically stationary box would remove the ambiguity of “when” altogether, since the share would become a time average with a standard error. It is the natural next experiment, with its own decision rule fixed before it runs.

VI. CONCLUSIONS

We isolated what no-slip walls do to decaying MHD turbulence by solving the Taylor–Green MHD flow in its closed fundamental box with free-slip and no-slip walls and nothing else changed, and we fixed the observable and the decision rule before the runs. No-slip walls add dissipation in a layer $\text{Re}^{-1/2}$ thick at every Reynolds number from 250 to 2000. The excess is viscous and comes with less Ohmic dissipation at the wall, and the wall layer intensifies as Re grows. A budget of the field’s energy, tested against a second rule fixed before its runs, attributes the Ohmic deficit to the stretching that the no-slip wall removes: within δ of a free-slip wall the flow feeds the field, and within δ of a no-slip wall the field does work on the flow. A smooth start, matched initial energy, a lower Mach number and $\text{Pm} = 0.5$ and 2 leave these findings standing, the start changing the size of the excess by up to 20% at $\text{Re} = 500$. Snapshots of the local dissipation show the same contrast in space: current sheets cross the near-wall layer of the free-slip box, and a broad viscous layer, with weaker currents, lines the no-slip walls. The pre-registered reading, the excess at the dissipation peak, falls between $\text{Re} = 1000$ and 2000, which is the outcome we had called “cascade”. It does so because the peak switches from one maximum of the dissipation to another. Over this range of Re the trend of the wall share depends on when it is measured, and its sign does not.

ACKNOWLEDGMENTS

The computations used the Computational Shared Facility (CSF3) of The University of Manchester.

AUTHOR DECLARATIONS

Conflict of interest

The author has no conflicts to disclose.

SUPPLEMENTARY MATERIAL

The supplementary material gives the tests of the method and the comparison with run C2 in full, the record of when each rule of the protocol was fixed, the failure of the first estimator of f_m , the resolution gate and controls rung by rung, the readings within 2δ and 4δ , the insulating pair, the fields at $t = 4.6$, the energy budget’s terms, checks and results rung by rung, and the four controls.

DATA AVAILABILITY

The code (both implementations), the job scripts with the decision rules fixed before the runs, the time series and wall-distance profiles of every run, the plane means of the field snapshots, the energy budgets, the analysis tools, the script that produces every number in this paper, and GitHub's record of when the protocol's commits were pushed are in the public M3LB repository, <https://github.com/alexderosis/M3LB>. The field dumps behind Figs. 4 and 5 and Figs. S1 and S2 (11 GB per run) are available from the author.

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